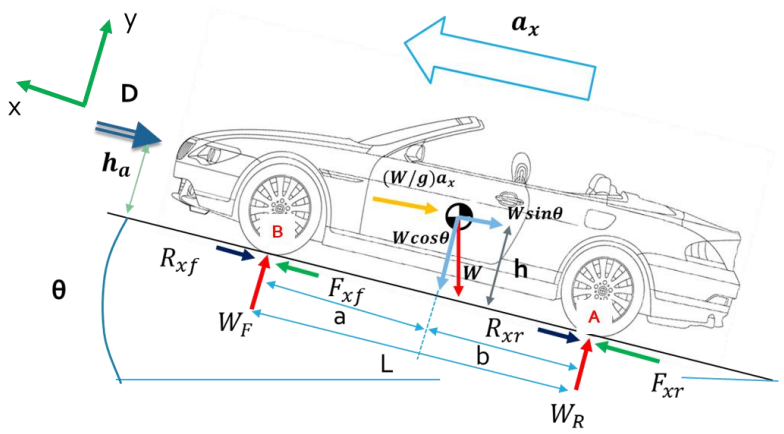
Longitudinal dynamics - Dynamic Force analysis of a Vehicle in Acceleration – Problem 2

An All-wheel drive car is accelerating at 10 m/s^2 with initial velocity of 40 kmph and it has a weight of 2000 kg , wheelbase is 2500 mm
Rolling resistance coefficient is 0.0150 , Drag coefficient * cross sectional area ; $C_D A$ = 0.554 m^2
It is travelling on an incline of 5 degrees
Find the Traction force required to maintain the acceleration if the traction is distributed evenly at front and rear wheels
CG height is 400 mm from ground and its located at 1400 mm with respect to the front wheel in x axis – notation "a" ,
Height of Drag force point is 400 mm from ground



$$h_a = 400\ mm\ from\ ground$$
$$h = 400\ mm\ from\ ground$$
$$C_{rr} = 0.0150$$

$$F_{xf} + F_{xr} = \left(+\left(\frac{W}{g}\right)(a_x)\right) + R_{xf} + R_{xr} + W\sin\theta + D$$

$$2F_x = \left(+(2000)(10)\right) + 0.0150 * W_f + 0.0150 * W_r + (2000 * 9.81)\sin\theta + D$$

Calculating Aerodynamic Drag force

$velocity\ 40kmph\ \ in \frac{m}{s} = 11.111$ $\quad$ $Drag\ area, C_D A = 0.554\ m^2$ $\quad$ $Density\ of\ air\ in\ kg/m^3 = 1.2754$

$$\textbf{Full equation}\ , D\ (in\ N) = \frac{1}{2}\rho v^2 C_D A$$

$$D = 43\ \mathrm{N}$$

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calculating Wf and Wr

$$W_f = \frac{-Dh_A - \left(\frac{W}{g} a_x h\right) - W sin\theta h + W cos\theta b}{L}$$

$$W_r = \frac{+Dh_A + \left(\frac{W}{g} a_x h\right) + W sin\theta h + W cos\theta a}{L}$$

$$W_f = \frac{-43 * 0.4 - (2000 \ * 10 * 0.4) - (19620)\sin(5)\,0.4 + (19620)\cos(5)\,1.1}{2.5}$$

$$W_f = 5119.47\ N$$

$$W_r = \frac{+43 * 0.4 + (2000 \ * 10 * 0.4) + (19620)\sin(5)\,0.4 + (19620)\cos(5)\,1.4}{2.5}$$

$$W_r = 14425.87\ N$$

$Substitute\ W_f \& W_r$

$$2F_x = \big(+(2000)(10)\big) + 0.0150 * W_f + 0.0150 * W_r + (2000 * 9.81)sin\theta + 43$$

$$2F_x = \big(+(2000)(10)\big) + 0.0150 * (5119.47) + 0.0150 * (14425.87) + (2000 * 9.81)sin\theta + 43$$

$$F_x = 11023\ N$$

Traction Force required at Front axle and Rear axle , $F_x = 11023\ N$